

Technical and geographic potential analysis for large-scale wind energy production using a GIS-AHP multi-criteria in Cameroon

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ABSTRACT

This article presents the suitability analysis of onshore wind farm sites using an AHP (Hierarchical Analysis Process) multi-criteria decision-making (MCDM) approach based on geographic information system modeling (GIS). It is based on different technical aspects (wind speed, elevation, slope), economic (proximity to the electricity network, proximity to roads) and social (proximity to residential areas, population density). The introduction of the factor “population density” and the constraint “required area” rarely used in previous studies makes it possible to refine the results. The data is classified into several categories in a gradual manner, namely unsuitable (0); less suitable (1); suitable (2); highly suitable (3) and most suitable (4). The results show us that the largest percentage relating to spatial coverage is observed in the “suitable” range with 50.89 % and the corresponding theoretical wind power potential (TWPP) is 1377.926 GW; it is followed by “highly suitable” 5.67 % and a TWPP of 153.652 GW; then “less suitable” with 1.04 % and a TWPP of 28.163 GW. No areas were identified in “most suitable” and finally the “unsuitable” range with 42.4 %. Also, the most favorable areas are located in the part of the far north of the country corresponding to the strong wind zone. The sensitivity analysis was carried out based on scenario tests (technical, economic and equal-weight) in order to give greater visibility of choice to the different stakeholders who would like to invest in onshore wind energy in Cameroon. Furthermore, a study of hybrid solar-wind systems could improve the efficiency of electricity parks in Cameroon.

1. Introduction

Cameroon, like most developing countries in sub-Saharan Africa, faces the challenge of generating more electricity to meet existing and future demand in a sustainable way. Despite the country's enormous renewable energy potential, its electricity sector remains underdeveloped and faces major financial difficulties, which have been compounded by the consequences of the COVID-19 pandemic and the war in Ukraine [1]. Nationwide access to electricity in Cameroon was around 65.4 % in 2021, very high in urban areas (around 94.5 %), in contrast to rural areas with around 24.8 % [2]. Although electricity production in Cameroon is essentially hydroelectric (around 76.1 %), peak load

management and the electrification of isolated areas require the use of thermal power plants that run on fossil fuels, namely oil and gas (around 23.9 %) [2]. Apart from the ever-increasing cost of these fossil fuels, their use emits greenhouse gases that are responsible for climate change. The war in Ukraine triggered a global energy crisis that led to soaring gas and electricity prices. In emerging and developing countries, the poorest households consume around nine times less energy than the richest, but spend a much higher proportion of their income on energy [1]. According to the International Energy Agency (IEA), global demand for electricity is set to increase by 25–30 % between now and 2030. Renewable energies, driven by solar photovoltaic and wind power, will play a crucial role in this energy transition [3]. This war is a major

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challenge for the energy sector, as it could lead to far-reaching, long-term policy changes aimed at accelerating the transition to renewable energies, which are more sustainable and secure energy systems. Investment in fossil fuels will inevitably decline, while investment in renewable energies will increase considerably [4]. This will lead to a reduction in the CO₂ emissions that are causing climate change.

Cameroon has a significant wind energy potential [5], which is favorable to electricity production under nominal wind turbine operating conditions [6,7]. The global climate emergency and the issues surrounding fossil fuels have prompted the development of sustainable energy production policies and strategies such as wind power [8]. As in many sub-Saharan countries, the development and use of wind power is hampered by the lack of data that would be of interest to decision-makers, the energy industry and others in terms of feasibility [9]. For a wind power project to be sustainable, techno-economic and socio-environmental assessment and appropriate planning are required. Several scientific investigations reported in the literature on wind energy in Cameroon are based on aspects such as the assessment of wind potential via various statistical techniques [10–12], and its use for energy production for various applications [13–16].

Furthermore, in order to propose suitable sites for wind farms in the regions of Cameroon, Metegam et al. [17] proposed a Boolean decision-making approach coupled with geographic information systems (GIS) using criteria such as slope, wind resource, proximity to the power grid, optimal distance to airports and accessibility by road, which are linked to geographical, climatic, economic and environmental aspects. This Boolean method coupled with GIS, while providing results, remains incomplete as it is based on a binary approach (True or False). The use of an MDCM-AHP method combined with GIS to calculate the different weights of the criteria will enable an optimal and more robust analysis of the problem.

The literature presents a taxonomy of studies based on multi-criteria decision models coupled to GIS (MDCM-GIS) in the case of proposing

suitable areas for the siting of wind power projects at different scales using various criteria [7,21–23]. To identify and select suitable sites for wind power generation in Varzeghan city in East Azarbaijan province, Asadi et al. [24] applied an innovative approach involving AHP-GIS-based linear regression modeling to five criteria (Distance from roads and network, Wind power density, Land slope, Distance from power lines, Distance from urban and protected areas). Also in the case of Sudan, Zalhaf et al. [25] applied a high-resolution GIS model coupled with Fuzzy-AHP (FAHP) to the same criteria as Islam et al. [7] for wind farm suitability analysis. Several other studies along the same lines have been carried out with multiple systems coupling MCDM to GIS [26–36]. Combining a GIS-based model and Type-2 FAHP with six criteria (Proximity to roads, Proximity to gridlines, Proximity to towns, Wind speed, Slope and Elevation) applied to Nigeria, Oyodele et al. [37] found that the best sites for wind farm development are located in the North. Watson and Hudson [38] in assessing the suitability of wind and solar farms over a large area of the south using multi-criteria evaluation coupled with GIS, found that suitability for wind power is generally low with only 0.5 km² representing the most suitable category. Jahangiri et al. [39] using the Boolean method of GIS software to identify ideal locations for the construction of wind power plants in the Middle East, concluded that the results found could be useful in creating sustainable energy development perspectives for natural resource-based systems, facilitating national strategies in the energy transport and environmental sector. Table 1 provides a summary of studies using combined GIS and decision-making for wind farm site selection, and Table 2 shows the decision criteria considered in previous studies of wind power plants. It can be seen that the most frequently used criteria are Proximity from roads, Wind speed, Proximity from gridlines, Slope, Proximity from residential areas. However, the criteria of population density and required site size are not used in most of these studies, yet they are essential in the identification of sites for large-scale grid-connected onshore wind farms.

Table 1
Studies using combined GIS and decision making for wind site selection.

	Authors	Year	Wind-Solar-power technologies	Criteria	Case study	Methods
1	Janke [50]	2010	Wind and Solar	8	USA	Multi-criteria GIS modelling
2	Jun et al. [51]	2014	Wind and Solar	13	China	ELECTRE-II
3	Watson and Hudson [52]	2015	Wind and Solar	7	UK	GIS and AHP
4	Mehdi J. et al. [39]	2016	Wind and Solar	/	Middle-East	GIS and Boolean
5	Jangid J. et al. [19]	2016	Wind	5	India	GIS and MCDM
6	Mohammad A. et al. [53]	2016	Solar and Wind	9	Afghanistan	GIS and MCDM
7	Baseer M.A. et al. [45]	2017	Wind	7	Saudi Arabia	GIS and AHP
8	Villacreses G. et al. [32]	2017	Wind	9	Ecuador	GIS and MCDM
9	Ayodele T.R. et al. [46]	2018	Wind	6	Nigeria	GIS and Fuzzy and AHP
10	Saeid M. et al. [47]	2018	Wind	16	Iran	GIS and MCDM
11	Kenji S. et al. [48]	2019	Wind and Solar	/	Bangladesh	GIS and MCDM
12	Shahid A. et al. [26]	2019	Wind and Solar	12	Thailand	GIS and AHP
13	Pasalari H. et al. [49]	2019	Wind and Solar	15	Shiraz city, Iran	GIS-FAHP
14	Koc A. et al. [31]	2019	Wind and Solar	7	Igdir Province/Turkey Ahmet	GIS and AHP
15	Moradi P. et al. [30]	2020	Wind	6	Alborz Province, Iran	GIS and AHP
16	Konstantinos I. et al. [29]	2020	Wind	5	Eastern Macedonia and Thrace region, Greece Ioannou	AHP and TOPSIS
17	Othman I. and Hushari M [23].	2020	Wind	5	Syria	GIS and AHP
20	Saraswat S.K. et al. [44]	2021	Wind and Solar	13	India	GIS and AHP
21	Fotsing I.F.M et al. [17]	2021	Wind	11	Cameroon	GIS-Boolean
22	Eroğlu H [27].	2021	Wind	17	Gümüşhane in Turkey	GIS-FAHP
23	Olivero V. et al. [28]	2021	Wind and Solar	Santa Marta, Colombia	GIS-AHP	
24	Suhrabuddin et al. [6]	2021	Wind	5	Herat, Afghanistan	GIS-FAHP
25	Rabiul Md. et al. [7]	2022	Wind	8	Bangladesh	GIS-AHP
26	Zalhaf A.S. et al. [25]	2022	Wind	8	Sudan	GIS-FAHP
27	Obaid S.A. and Anzah F [21].	2023	Wind and Solar	5	Kuwaiti desert	GIS-AHP
28	Tarife R. et al. [22]	2023	Wind, Solar and Hydro	9	Southern Philippines	GIS-FAHP
29	Meysam A. et al. [24]	2023	Wind and Solar	5	East Azarbaijan province	GIS-AHP and Linear Regression Model

Table 2
Decision criteria considered in the previous studies wind plants [40,56,57].

[31]	[23]	[30]	[29]	[6]	[7]	[25]	[21]	[22]	[24]	[17]	[57]	[56]	[33]	[46]	[55]	[44]	[45]	[40]	[54]	Criteria
																				Wind resources (wind speed)
																				Slope
																				Aspect
																				Elevation
																				Distance from Coastline
																				Distance from waterbodies
																				Distance from airports
																				Distance from wildlife
																				land-use
																				Distance from residential area
																				Distance from roads
																				Distance from transmission lines
																				Distance from power plants
																				Distance from telecommunications
																				Distance from tourist facilities
																				Population density
																				Farm required area
																				Birds area

The aim of this study is to evaluate the technical and geographical potential of large-scale wind farm sites that can be connected to the national grid, using a multi-criteria decision-making tool coupled with a geographic information system (MDCM-AHP-GIS). Population density, the constraint allowing the exclusion of sites smaller than 4 km² and new criteria introduced help to refine the results and optimize site selection and standards in onshore. In this study, the various suggestions made by regional experts, stakeholders, experts from different environmental associations, energy practitioners and planning authorities are considered in order to make the results more meaningful. The approach proposed for wind energy is the most innovative in the literature in the case of Cameroon.

The specificity of this study stems from the use of GIS-AHP in Cameroon, a nation in sub-Saharan Africa where such research is unprecedented. Incorporating new factors like population density and the land needed for wind farms linked to the current power grid aids in refining site selection and enhancing network profitability. It's crucial to note that renewable energy sources, especially wind energy, are nascent in Cameroon. Hence, this research could serve as a vital resource for investors seeking viable investment opportunities.

2. Materials and methodology

In this study, a multi-criteria decision analysis (MCDM) method

coupled with GIS was adopted in order to optimally propose the selection of regions as suitable wind power plant siting sites in the Cameroon context. The main factors playing a major role in the appropriate siting of a wind power plant are identified and in common agreement with those derived from the literature (Table 2). The AHP method was used to calculate the corresponding weights for each criterion, taking into account their relative importance assigned on the basis of the matrix completed by the experts. Data for this study were collected at different scales and all resampled to the exact raster resolution (1 km × 1 km). Vector data were converted to raster format in order to homogenize all study parameters. The constraints and factors used are presented in Tables 4 and 5.

2.1. Study area

Cameroon is a Central African country with an area of 475,650 km² and an estimated population of over 25 million in 2019 (Fig. 1). It has ten regions and a variety of climates across its territory, with a 420 km maritime border along the Atlantic Ocean in the southwest [58]. According to the literature [10–18], there are significant reservoirs of wind generation in the areas of Kaelé and Kousséri in the Far North, the Bamboutos Mountains in the West region and certain localities in the Adamaoua region, Centre and Douala, Limbe, Kribi in the Littoral. Increasing demand for electricity due to population growth,

Table 3

List of data layers, their types and sources.

Data layer	Types (format)	Resolution	Geometry	Sources
Administrative limits of Cameroon (regions, departments, districts) Map	Vector (shapefile)	–	polygon	GADM, 2022 [64]
Wind speed m/s at 100m	Raster	(1*1 km ²)	Global Wind Atlas [60]	
Map of population density in Cameroon	Raster	(1*1ha)	Wordpop, 2010 [61]	
Map of Cameroon Power lines	Vector (shapefile)	–	Point	World Bank [65]
Hydrological map of Cameroon (streams, navigable waters, rivers, rivers, wetlands, reservoirs ...).	Vector (shapefile)	–	Line, polygon	OSM, 2022 [63]
Map of land use in Cameroon Map	Vector (shapefile)	–	polygon	OSM, 2022 [63]
Map of the road network (inter_state, primary, secondary roads ...) in Cameroon.	Vector (shapefile)	–	lines	OSM, 2022 [63]
Map of elevation and slope in Cameroon	Raster	(1*1 km ²)	Global Wind Atlas [71]	
Map of Cameroon airport	Vector (text format csv) Geometry	–	Point	ADC (Cameroon airport), 2022 [62]

Table 4

Classical AHP adapted from Saaty [90].

AHP Scale	Definition
1	Equally Important
3	Weakly important
5	Fairly important
7	Strong important
9	Absolutely important
2, 4, 6, 8	Intermediate value between above scale values

socio-economic development and other factors have led to the development of several wind power investment projects currently under study in Cameroon (Monts Bamboutos, in the West, North-West and South-West regions, with capacities of 42 MW, expandable to 80 MW).

2.2. Data sources

The data required for this study was collected and evaluated prior to analysis in a GIS environment. The portals of various governmental and international organizations were consulted to obtain the latest available data. Data on wind speed at 100m height, altitude and slope were collected from the Global Wind Atlas site at a spatial resolution of 1 km² [59,60]. Population density data are obtained from WorldPop with a

Table 5

Random index (RI).

n	1	2	3	4	5	6	7	8	9	10
RI	0.00	0.00	0.058	0.90	1.12	1.24	1.32	1.41	1.45	1.49

resolution of 1*1ha [77]. Airport location data are obtained from the Cameroon ADC airport site [62]. Land use and land cover data are obtained from OpenStreetMap [63]. Data on administrative boundaries are obtained from GADM [64]. Data on power lines are obtained from the World Bank website [65]. The description of the data used in this study is listed in Table 3, together with the type of format used, the resolution and their various sources, accompanied by Fig. 3. Wind speed data are taken from the DTU Global Wind Atlas Project [60].

2.3. Analysis of wind resource siting criteria

The choice of the various selection criteria for this study was finalized after an in-depth review of the literature, and complies with national, environmental and international guidelines. These criteria were also validated by 10 experts whose profiles are listed in Table 9. This was done in order to minimize any conflicts of interest or personal biases [66]. Eight of the selected experts are based in Cameroon and two in Belgium, and are university professors, professional engineers and researchers with sound knowledge of wind energy applications. The experts based in Cameroon have a good knowledge of the soil conditions in the study area.

Although wind speed (m/s) is an essential parameter to consider when choosing a large-scale wind farm connected to the electricity grid, technical-economic and socio-environmental factors are also important when choosing a wind farm [20,22,24]. Added to this are climatic factors, topography and interactions with the surrounding environment that can generate negative externalities [30]. Seven factors are taken into account when selecting sites for wind farms: wind speed, elevation, slope, distance from residential areas, distance from power lines, population density and distance from roads.

2.3.1. Wind speed

Being one of the factors in the choice of favorable sites, it can be calculated from Equation (1) as a function of height with $n = 0.4$ the friction coefficient or surface roughness, V_2 is the wind speed at height Z_2 , V_1 is the wind speed at reference height Z_1 [67].

$$V_2 = V_1 \left(\frac{Z_2}{Z_1} \right)^n \quad (1)$$

The wind energy density per unit area can be obtained from Equation (2) where P (in W) represents the average wind power, ρ the air density (1.225 kg/m³ for the present case), A (in m²) the swept area [67–69].

$$\frac{P}{A} = \frac{1}{2} \rho V^3 \quad (2)$$

Several authors have considered wind speed as the most important factor for a wind farm (Table 1). The minimum wind speed required to start production is 3 m/s [19]. Several studies have been carried out with different wind speeds, such as Ali et al. [26], which considered a minimum wind speed of 4 m/s. In this study, the considerations are as follows: “unsuitable” when the average wind speed is below 4 m/s, “less suitable” when the speed is between 4 and 5 m/s, “suitable” when the speed is between 5 and 6 m/s, “highly suitable” when the wind speed is between 6 and 7 m/s, and “most suitable” when the wind speed is above 7 m/s. GRASS GIS 7 was used to assemble the final raster file. Fig. 3-a shows the spatial distribution of Cameroon’s mean annual wind speed at 100m height. It can be seen from this figure that the lowest wind speed in the study area is 0.4 m/s and the highest is 11 m/s.

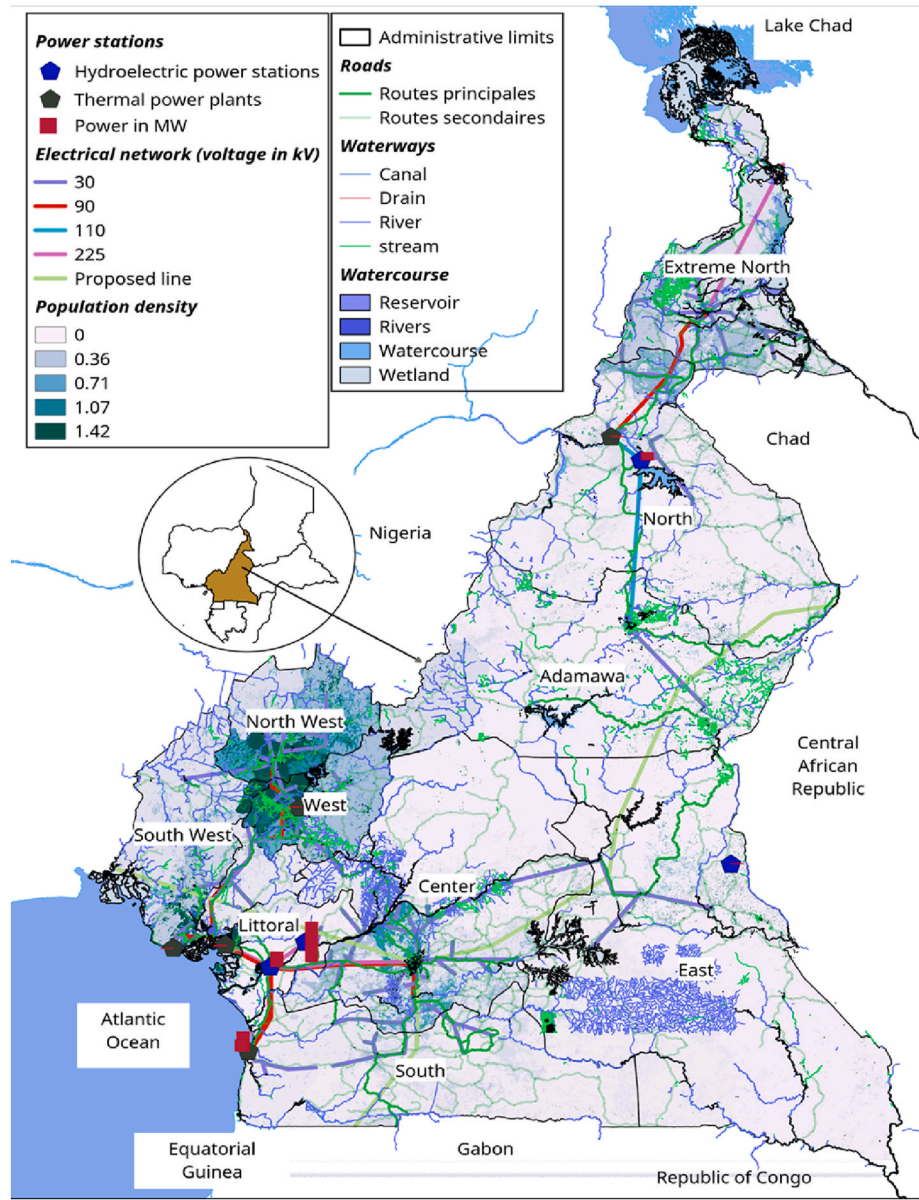


Fig. 1. Map of Cameroon's energy infrastructure.

Table 6
Restriction criteria from chosen wind farm site selection studies.

Plant required area in km ² [93]	Wind speed (m/s)/ density Potential [9]	Buffer distance/ proximity from electricity grid, m [9]	Proximity/ buffer to roads & highways, m [94];	Buffer distance from forests & parks, m	Buffer distance from airports, m [94];	Buffer distance/ proximity from residential, m [94];	Buffer distance from lakes, m [94]	Buffer distance from rivers, m [94]	Slope, % [94]	Elevation (m) [9]
≥4	>4	>250 and <10000	>500 and <10000	>300	>3500	>2000	>400	>400	<10	<2000

2.3.2. Distance from the power grid

Several authors have used proximity to the power grid as an important parameter for the siting of large-scale wind farms connected to the existing power grid (Table 2). The investment cost of a grid-connected wind farm construction project is reduced when electricity is delivered via existing transmission lines. On the other hand, long transmission lines between wind farms and the power grid are associated with cabling costs and electricity losses. For safety reasons, a minimum distance of 100 m will be observed between the wind farm and

overhead power lines to minimize damage caused by any projections made by the wind turbines [54]. For the purposes of this study, a distance of 10 km around the power plant is considered favorable for the feasibility of wind power plants. Information about the power plant and maps of the national grid were obtained from the company in charge of distributing and marketing electricity in Cameroon [26]. Beyond 10 km, the area is considered unsuitable. Data on the electricity grid comes from the World Bank 2022. Fig. 3-f shows the spatial distribution of Cameroon's high-voltage electricity network.

Table 7

Reclassification of weighted criteria for wind power [9].

Category	Economic			Technical			Social	score	Relevance
	Proximity (m)			Wind speed (m/s) at 100m	Slope (%)	Elevation (m)	Population density, Minimize density in inhabitants/km ²		
	Proximity from roads and highways	Proximity from electricity grid(in m)	proximity from residential,						
A	<500	<250	<2000	<4	>15	2001–2384	>500	0	Unsuitable
B	>15000	>20000	2001–6000	4–5	10–15	1001–2000	500–100	1	less suitable
C	10001–15000	10001–20000	6001–10000	5–6	6–10	501–1000	50–100	2	suitable
D	5001–10000	5001–10000	10001–20000	6–7	3–6	201–500	1–50	3	Highly suitable
E	501–5000	251–5000	>20000	>7	<3	<200	0	4	Most suitable
Sensitivity analysis	0.104	0.147	0.096	0.326	0.043	0.082	0.224	weight	
	10 %	15 %	9 %	32 %	4 %	8 %	22 %	Normalized weight	
	34 %			44 %			22 %	CR = 8 %	
	6.5 %	15 %	5 %	35 %	10 %	15.7 %	12.8 %	Scenario 1 (Technical weight)	
								CR = 4.5 %	
	26.5 %			60.7 %			12.8 %	Scenario 2 (Economic weight)	
	15.7 %	35 %	10 %	15 %	5 %	6.5 %	12.8 %	CR = 4.5 %	
								Scenario 3 (Equal Weight)	
	60.7 %			26.5 %			12.8 %		
	14 %	15 %	14 %	15 %	14 %	14 %	14 %		
	43 %			43 %			14 %		

Table 8

Statistical information of land suitability areas for three sensitivity cases.

Range	Categories	Statistics	AHP-Wind	TWPP is the theoretical wind power potential (GW)	Scenario 1 (Technical)	TWPP is the theoretical wind power potential (GW) on scenario 1	Scenario 2 (Economic)	TWPP is the theoretical wind power potential (GW) on scenario 2	Scenario 3 (Equal weights)	TWPP is the theoretical wind power potential (GW) on scenario 3
0–0	0	Areas in km ²	243147.935	243147.935	4.722	243147.935	243147.935			
		Pixels number	24313443		24313443		24313443		24313443	
		%	42.40 %		42.40 %		42.40 %		42.40 %	
0.001–1	1	Areas in km ²	5964.161	28.163	34971.342	165.135	3882.816	18.335	9312.7672	43.975
		Pixels number	596383		3496940		388260		931225	
		%	1.04 %		6.10 %		0.68 %		1.62 %	
1.001–2	2	Areas in km ²	291809.767	1377.926	248115.271	1171.6	204020.342	963.384	240502.028	1135.651
		Pixels number	29179356		24810149		20400901		24048867	
		%	50.89 %		43.26 %		35.58 %		41.94 %	
2.001–3	3	Areas in km ²	32539.747	153.652	46824.981	221.108	108646.7543	513.03	80518.772	380.21
		Pixels number	3253794		4682238		10864072		8051430	
		%	5.67 %		8.17 %		18.95 %		14.04 %	
3.001–4	4	Areas in km ²	0	0	423.264	2	13784.946	65.093	1.29	6.091
		Pixels number	0		42324		1378418		129	
		%	0 %		0.07 %		2.40 %		0.00001 %	

2.3.3. Distance from residential areas

Proximity to residential areas is a very important factor used by several authors for the siting of large-scale wind farms connected to the existing electricity grid (Table 2). The siting of wind farms close to residential areas has negative social and environmental implications. The negative effects observed can be light reflections, flickering shadows or landscape effects. Accidents caused by the falling propellers of damaged wind turbines, and noise nuisance caused by.

The rotation of wind turbines have also been observed. Although there is no worldwide agreement on the appropriate distances of wind turbines from dwellings, it is nevertheless important to have a minimum safety distance between wind farms and residential areas in order to reduce socio-environmental impacts. Based on the literature review and international wind energy guidelines from several countries, large-scale wind farms should be built far away from residential areas [70,73]. Furthermore, it is important that wind farm sites are located at a

Table 9

List of experts.

N°	Designation	Qualification	Age (in year)	Work experience (in year)	Department/ Company
1	Professor	PhD	36	12	University of Dschang, Cameroon (UDs)
2	Professor	PhD	50	25	University of Dschang, Cameroon (UDs)
3	Lecturer	Graduate	55	25	Free University of Brussels (ULB)
4	Energy expert	Graduate	48	23	Ministry of energy, Cameroon
5	Energy expert	Graduate	42	18	Ministry of energy, Cameroon
6	Deputy- Manager	PhD	38	8	Solar Energy Technology, Cameroon
7	Deputy- Manager	Graduate	37	11	Instrumelec, Cameroon
8	Lecturer	PhD	52	22	Free University of Brussels (ULB)
9	Assistant- Manager	Graduate	30	7	ENEO, Cameroon
10	Assistant- Manager	Graduate	35	11	SONATREL, Cameroon

reasonable distance from residential areas to minimize transmission losses. In this study, a safety distance of 2 km around residential areas was taken in order to reduce socio-environmental impacts as in Refs. [53,72]. Data on residential areas are obtained from Open Street Map (OSM 2022) [63]. Fig. 3-h shows the spatial distribution of residential areas in Cameroon. It can be observed that the East Cameroon zone is the most sparsely populated. This is due to the forest that covers a large part of this region.

2.3.4. Distance from roads

Large-scale wind farms must be accessible from the existing road network to avoid the need to build new roads to serve the site. In addition, expenditure on site construction and maintenance will be considerably reduced if the site is easily accessible by vehicles for transporting materials. According to Table 2, proximity to road networks has been used by several authors as a very important economic factor in reducing investment costs when siting large-scale wind farms. However, a safe distance must be maintained to reduce the risk of fatalities due to accidents caused by projected materials or falling turbine parts. On the basis of the literature review, it has been assumed a safety distance of 500 m in this study as Othman et al. [23] and Albraheem et al. [72]. To reduce transport costs, it is preferable to keep the distance from the road as short as possible. A distance of 10 km around roads is considered to be a suitable area for siting wind power plants, and areas greater than 10 km are considered inappropriate [26,41–43,53]. Road data is taken from Open Street Map 2022 [63]. Fig. 3-e shows the spatial distribution of the road network in Cameroon. The data have been filtered to group the main types of road (primary, secondary and tertiary). In this study, it has been used primary and secondary roads, as tertiary roads here are mainly tracks and are very poorly developed and maintained. The construction of a wind farm close to tertiary roads would require regular road development and maintenance. This would lead to additional costs.

2.3.5. Slope

Accessibility for the installation and maintenance of large-scale wind farms is hampered by steep slopes, as these prevent installation

equipment such as trucks and cranes from depositing the materials needed to erect the turbines on site. Steep slopes increase the cost of construction and maintenance of wind farm installations. For this reason, a maximum threshold should be observed when selecting a wind farm site. Furthermore, wind speed increases with height. It therefore makes sense to choose a slope that would optimize the site's profitability while also being economically viable. According to the literature, a variation between 10 % and 30 % is authorized as the maximum slope threshold and for this study the maximum threshold of 10 % is considered [33,53]. The slope is taken from the elevation map provided by the DTU Global Wind Atlas Project [60]. Fig. 3-b shows the spatial distribution of slope in Cameroon. This map was obtained from DEM using the spatial analysis tool (r.slope.aspect) in GRASS-GIS 7.4.3. It can be seen that the slope in Cameroon varies from 0 to 1.290 %. However, most of the study area has a slope of less than 9 %.

2.3.6. Elevation

Considered in the literature as one of the important factors in the judicious choice of sites for wind power plants, the lower altitude of a site is considered to be favorable for the siting of a wind power plant [7, 33,34]. However, the opposite is less appreciated due to the additional costs involved. For this study, according to Sadeghi et al. [34] and Saraswat et al. [44], an altitude of 2 km is considered the maximum threshold. The spatial distribution of altitude from USGS/NASA SRTM data is shown in Fig. 3-c.

2.3.7. Population density

Although population density is an important social parameter for the selection of renewable energy plants, to the best of our knowledge, no author has used it as a factor in the selection of the suitable site for the siting of large-scale wind power. However, it has been used by Yushchenko et al. [74], for site selection of photovoltaic (PV) and concentrating solar power (CSP) parks connected to the existing and unconnected electricity grid. For grid-connected sites, the authors minimized population density. Moreover, according to International Renewable Energy Agency (IRENA), population density is an important social parameter when choosing a renewable energy site [75]. Indeed, the negative impact of wind farms on the population means that they should be built as far away from densely populated areas as possible. In this study, areas with a population density of over 500 habitants/km² are considered unsuitable for wind farms [76]. Fig. 3-d shows the spatial distribution of population density in Cameroon. It can see that population density varies from 0 to 1424 inhabitants per km² in Cameroon. The most densely populated regions are the West and Littoral. The high population density observed in the Littoral region is due to the presence of the economic capital in this region.

2.3.8. Constraint factors

2.3.8.1. Airport safety distance. Wind turbines can seriously affect the processes of communication, navigation and surveillance systems used in air traffic control by interfering with the radar signal. Islam et al. [7] and Sadeghi et al. [34] have considered an important safety distance around airports. In this study, a buffer of 3500 m is observed around the various airports in Cameroon [53]. Relative data on the position of airports in Cameroon come from Cameroon Airport (ADC 2022) [62]. Fig. 3-i shows the geographical location of the eight airports in Cameroon.

2.3.8.2. Land use and protected areas. When considering the selection of suitable sites for large-scale onshore wind farms, several authors have excluded protected areas and land use from their study area (Table 2). Indeed, wind farms should not be located in forests, woodlands, protected areas, wildlife reserves, archaeological and historical sites, tourist areas and parks. According to studies carried out by INRA (National

Institute of Agricultural Research) in France and ANSES (National Agency for Food, Environmental and Occupational Health Safety), electromagnetic fields generated by high or medium-voltage lines, transformers, wind turbines or other electrical sources can create stray currents above a certain threshold, which can have a negative impact on animals and farms located nearby [77]. In this study, protected areas and land use were considered unsuitable for the installation of wind farms, and a 300m buffer zone was established around these areas to reduce environmental impacts [54]. Land use data are taken from OSM 2022 [63]. Fig. 3-h shows the spatial distribution of protected areas in Cameroon. They are grouped into four categories (parks, forests, residential and nature reserves). Table 6 presents the restriction criteria utilized in this study, while Fig. 2 displays the constraint map generated by applying these criteria in GRASS-GIS software. In Fig. 2, areas marked with “0” are unsuitable, and those marked with “1” are deemed suitable for locating large-scale wind farms connected to the electrical

grid.

2.3.8.3. Water bodies and wetlands. Some authors have used water bodies and wetlands as a parameter that can influence the installation of large-scale wind farms (Table 2). The construction of wind farms in water bodies or wetlands is not advisable, as it would block the flow of water and be more costly [23,72]. Consequently, a buffer of 400 m was taken as a safety distance around water bodies or wetlands [23,53,54]. Fig. 3-g show the Watercourses map of Cameroon.

2.3.8.4. Plant required. Some authors have used the minimum surface area required as an important constraint factor for the installation of a large-scale onshore wind farm connected to the existing electricity grid.

Indeed, in this study, it can be identify suitable sites for the construction of large-scale wind power plants connected to the electricity grid in order to produce the electrical energy required to supplement the

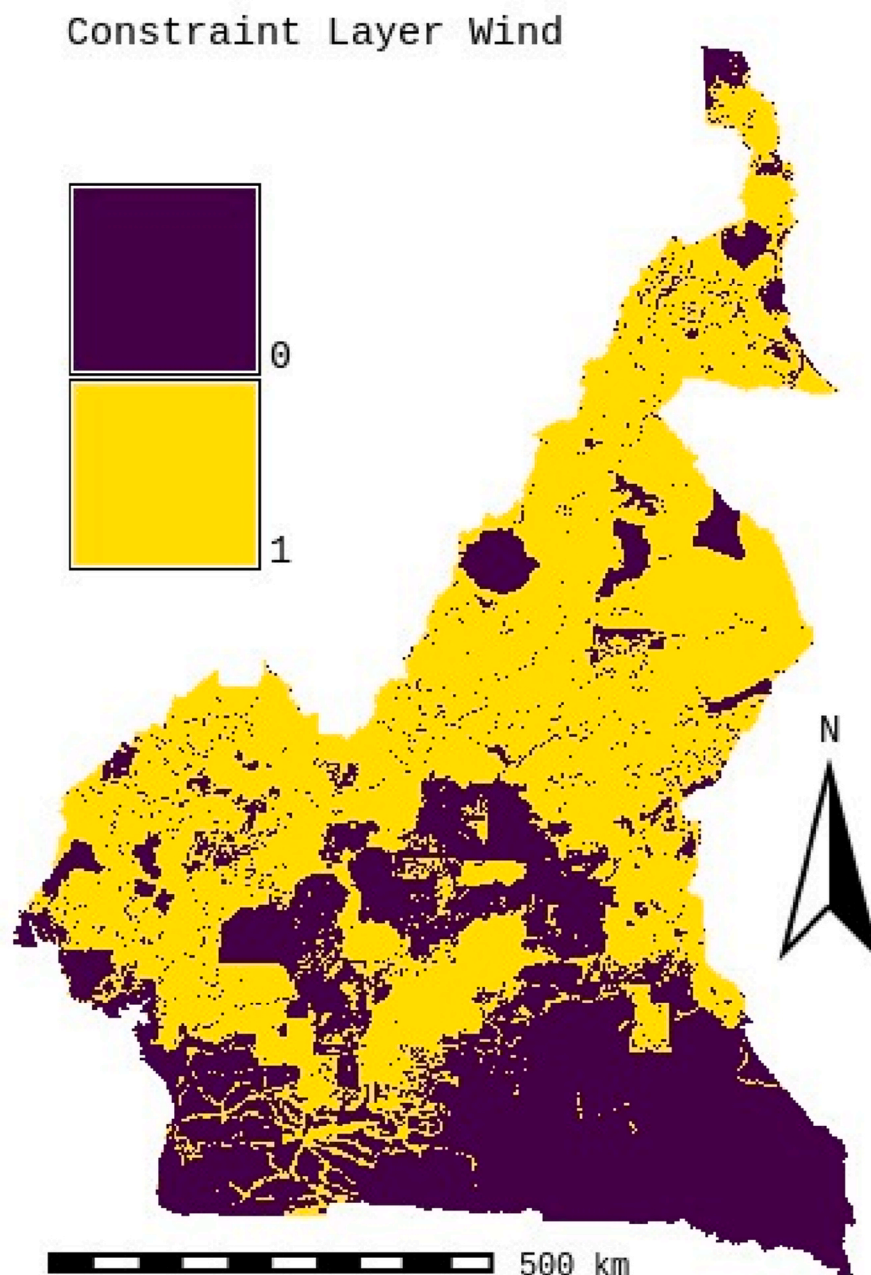


Fig. 2. Constraint layer (“0” is not suitable and “1” is suitable map).

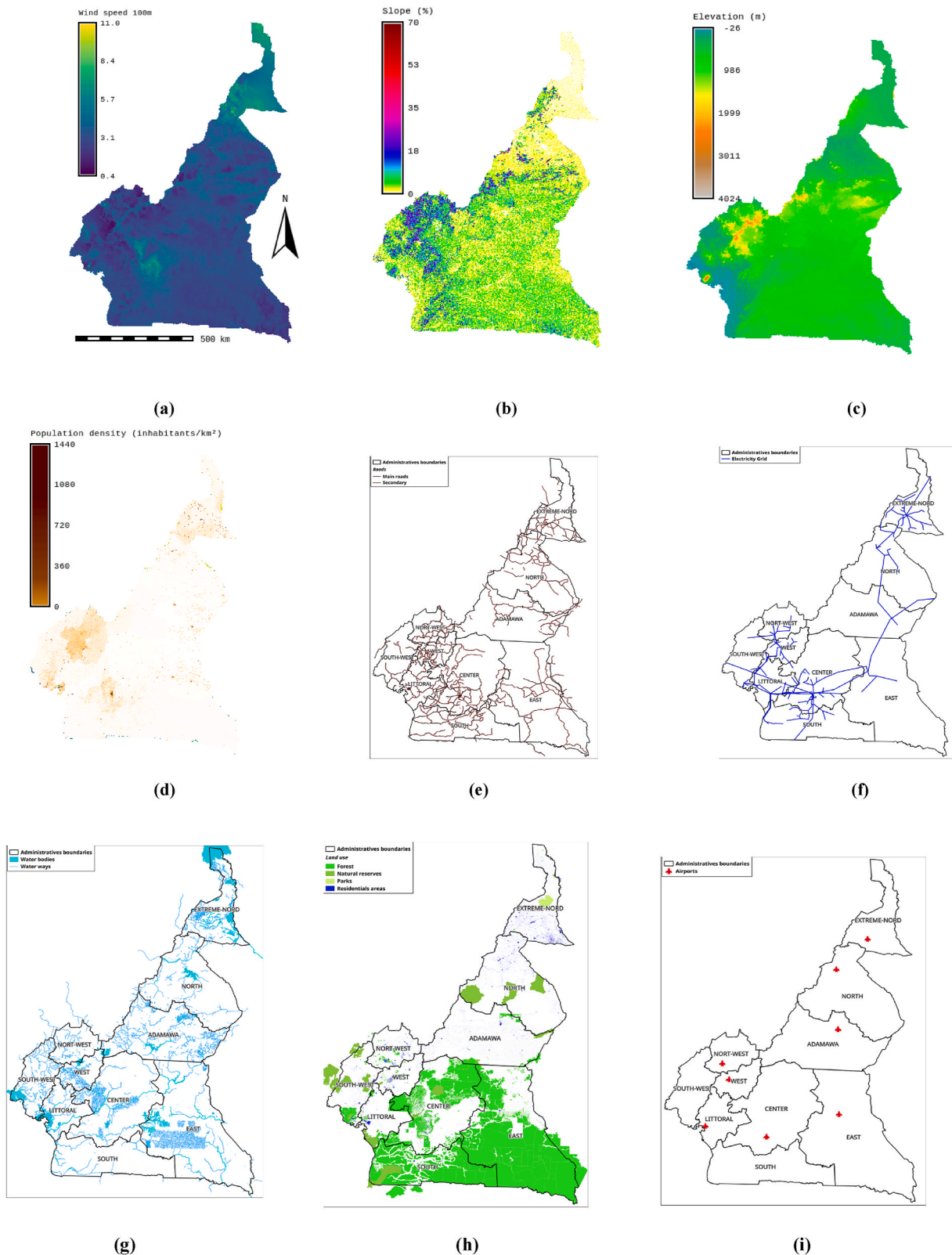


Fig. 3. Study area a) Wind speed at 100m, b) Slope Map, c) Elevation Map, d) Population density, e) Roads Map, f) Transmission network Map, g) Watercourses, h) Land cover Map, i) Airport Map.

population's electricity needs. Consequently, it is essential that these plants have the dimensions required to cover the population's electricity needs. In this study, the minimum area required for the installation of a wind farm is 4 km² [53,55]. The turbine used in this study to assess the

theoretical wind power potential is the V110-2.0 MW® IEC IIIA turbine. Fig. 3 shows the distribution of the various study inputs. These include the spatial distribution of the power grid, roads, watercourses, elevation, slope, landcover and landuse, population density, airports ...

2.4. Methodology

The methodological approach employed combines the Analytic Hierarchy Process (AHP) decision-making method with Geographic Information Systems (GIS), with details illustrated in Fig. 4.

2.4.1. Geographical information systems (GIS)

GIS have been used as a research and application tool since the 1970s, enabling the creation, management, analysis and mapping of all types of data and presenting a variety of applications [78–81]. In this study, all geographic information was processed in raster form at a resolution of 1 km*1 km. Vector data (waterbodies, power lines, road network, residential areas, spatial distribution of airports, land use ...) were rasterized to make them homogeneous with raster data (wind speed, elevation, population density ...). In this study, Boolean logic based on 0 for non-appropriate and 1 for appropriate is used to extract inappropriate areas. The free software GRASS GIS.7 and QGIS 2.18 were used to pixelate and normalize the data layers in this study. QGIS 2.18 was used to extract data from Cameroon and to create layouts. XLSTAT software was used to calculate the weights of the various parameters and the coherence index (CI). GRASS GIS.7 software was used to carry out the methodology and extract the results. Statistical analysis was also carried out in GRASS GIS.7 and presented using LibreOffice Calc.

2.4.2. MCDM-AHP method

The MCDM-AHP method proposed as a tool in this study is one of the multi-criteria decision methods used in several aspects including those of renewable and conventional energy planning. It offers a number of advantages, such as the ease with which site alternatives can be ranked, and the calculation of an aggregated synthetic score based on a hierarchical ranking and weighting of all the criteria taken into account in the decision [82–85]. Due to the limitations of the AHP linked to rank

inversion occurring when adding or deleting parameters influencing others, a consistency test is used to filter out inconsistent judgments [86, 87]. In addition, tools such as XLSTAT are used to overcome the influence of rank inversion. This presents AHP as a more robust, flexible method facilitating the resolution of complex decision problems [88], in line with its GIS-based use in recent years in several scientific works [7, 21,23,28,31,89–92,95].

It is appropriate to initially qualify the decision-making framework relating to the decision problem. This step is applicable regardless of the Multi-Criteria Decision Making (MCDM) method chosen and allows the establishment of the entity responsible for the decision, the limits of the evaluation, and to define in particular what are the development alternatives (or scenarios) to be evaluated, and what are the relevant criteria (technical, economic, social) to use to compare them.

The main stages of the AHP methodology are described below:

Step 1 Develop the hierarchical structure of the decision problem.

Step 2 Develop a pairwise comparison matrix of the decision problem as illustrated in Equation (3) using the Saaty point scale of 1–9 based on Table 4.

$$A_{ij} = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \dots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{pmatrix} \quad (3)$$

Step 3 Calculate the consistency index (CI) from Equation (4). The CI consistency index is used to measure the consistency of the pairwise comparison of the matrix.

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (4)$$

$\lambda_{\max}$ is the eigenvalue and n is the number of main criteria which is

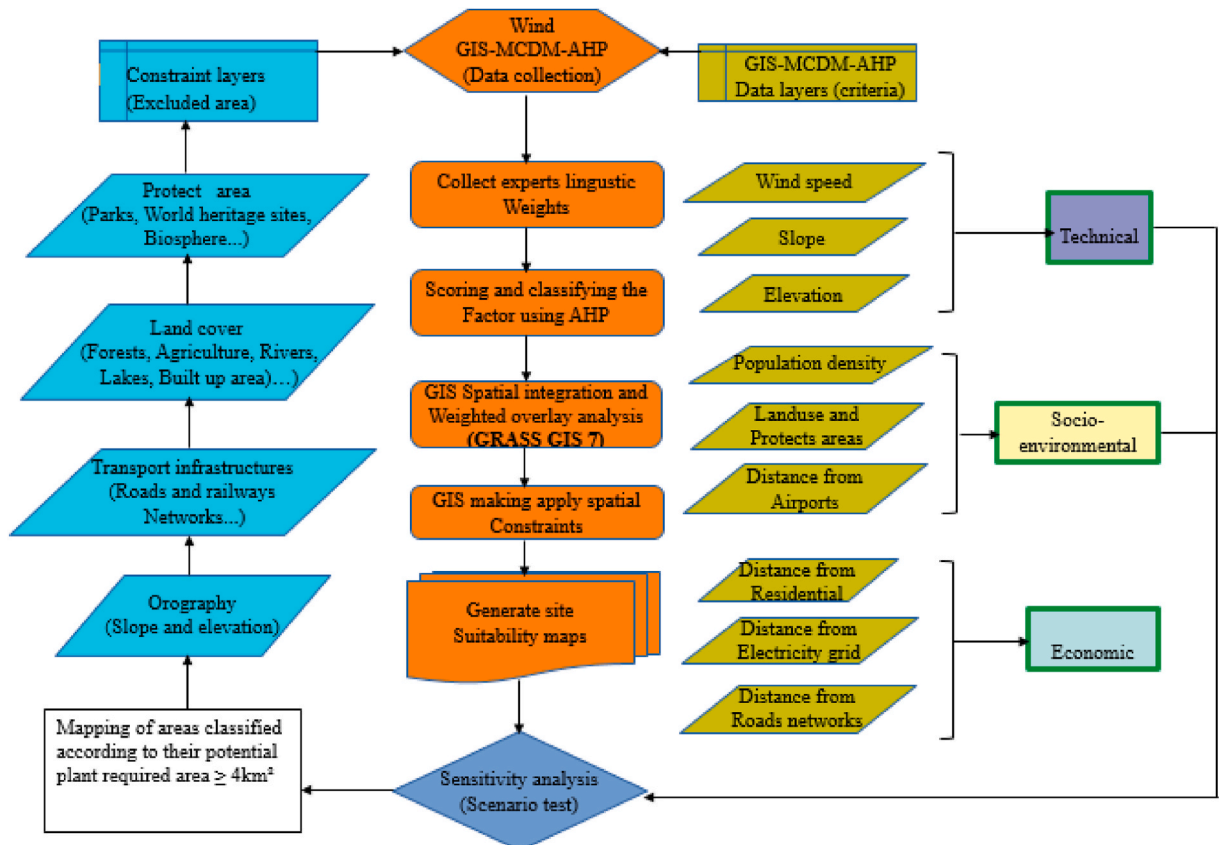


Fig. 4. Methodology of study.

seven in this study.

Step 4 Calculate the consistency rate (CR) from Equation (5)

$$CR = \frac{CI}{RI} \quad (5)$$

where RI is the random consistency index which depends on the number of main criteria as shown in Table 5. In this study, the RI is 1.32 because, the number of main criteria is seven. When comparing pairwise, the consistency rate (CR) must be less than 10 %; if it exceeds 10 %, the results may be inconsistent. In this study, the random consistency index CI is 8 %. For the economic and technical scenarios, the CI is 4.5 %. It can be therefore conclude that these values are adequate for the study. In this study, the AHP method allowed us to obtain the different weights of the 7 main criteria. The total fit score (S) is calculated by adding the factor weights according to Equation (6):

$$S = \sum_{i=1}^{i=N} W_i \cdot P_i \quad (6)$$

where W_i is the i th weight of the criterion and P_i the score of the criterion of the i th factor. Table 7 presents the factors and weightings chosen for the evaluation of wind energy production potential. The different weightings corresponding to the sensitivity analysis are also listed in this table. The final map is obtained by multiplying the map resulting from Equation (6) by the constraint layer using the “Raster calculator” tool in the free software GRASS GIS 7. The formula used is that presented in Equation (7).

$$\text{SuitabilityMap} = \left(\sum_{i=1}^{i=N} W_i \cdot P_i \right) * \text{constraint} \quad (7)$$

2.4.3. Theoretical wind power potential

The theoretical wind energy potential can be determined from Equation (8), where TA represents the total area of suitable land (in km²), AF represents the area factor in MW/km². The turbine used in this study to assess the theoretical wind power potential is the V110–2.0 MW® IEC IIIA turbine because it is extremely reliable and is documented by its high availability and performance [47,53]. This turbine helps increase productivity by opening up low wind sites that were previously considered unviable. To maximize power output at such locations, the wind turbines’ 110m rotor takes more advantage of the wind available from an incredibly low 3 m/s. With its 54m blades, the V110–2.0 MW® IEC IIIA offers an outstanding rotor-to-generator ratio producing outstanding capacity and efficiency in low to medium wind sites. The area factor of this turbine is 4.722 MW/km². This turbine was also used by Saraswat et al. [44] in India to assess India’s theoretical wind power potential.

$$TWPP = TA * AF \quad (8)$$

3. Results and discussion

3.1. Final determination of suitable lands

The analysis of wind energy production potential was based on the two scenarios presented in Table 7. To reclassify the various factors, land suitability indices were associated with the different scores, as follows: 0-Unsuitable, 1- Less suitable, 2-Suitable, 3- Highly suitable and 4- Most suitable.

Fig. 5 shows the maps corresponding to the reclassified layers of location factors for wind farm sites. Fig. 6-a displays the map generated by merging the maps from Fig. 5 using the “raster calculator” tool in GRASS-GIS software, applying the various weights derived from the AHP method for the initial scenario. Fig. 7-a, 7-b, and 7-c represent the final maps produced by combining the maps of Fig. 5 with the weights from Table 7, which pertain to the sensitivity analysis. These maps were

subsequently divided into four categories to determine the areas needed to calculate the theoretical electric potential. The categorized results are shown in Fig. 6-b, 8-a, 8-b, and 8-c. Fig. 9 shows for land suitability areas for three sensitivity cases. The analysis in Table 7 describes the ranking of aspects according to scores, with 22 % for the social aspect, 34 % for the economic aspect and 44 % for the technical aspect. Concerning the factors, it can be observed that the weight of wind speed is 32 %, population density with 22 %, distance to the power grid with 15 % of weight and finally distance to the road network with 10 % of weight.

This is followed by the factors “proximity to residential”, “elevation” and “slope” with weights of 9 %, 8 % and 4 % respectively. All these features are listed in Table 7. The scores and capability maps of the GIS-AHP combination are shown in Fig. 6-a and b. After analyzing Table 8 listing the statistical information and the theoretical wind potential, it can be seen that 42.40 % of Cameroonian territory is considered unsuitable for the installation of large-scale wind power plants. This same Table 8 shows that 5.67 % is most suitable with a wind potential of approximately 153.652 GW. 50.89 %; 1.04 % and 0 % are suitability respectively; less suitability and highly suitability with wind potentials of 1377.926 GW and 28.163 GW and 0 GW respectively.

3.2. Sensitivity analysis

The sensitivity analysis takes into account three scenarios: the technical scenario, which assigns a higher weight to the technical factors (scenario 1); the economic scenario, which assigns a higher weight to the economic factors (scenario 2); and the equal weight scenario (scenario 3), which assigns the same weight to all factors. Table 8 lists statistical information on land suitability zones and theoretical wind power potential in GW for the different scenarios.

The combined technical scenario (scenario 1) assigns the greatest weight to technical factors (60.7 % wind speed, elevation and slope) and the lowest weights to economic and social aspects (26.5 % and 12.8 % respectively). The weights of the factors corresponding to the technical scenario are listed in Table 7, and the suitability maps resulting from the GIS-AHP combination are illustrated in Fig. 5-a and 6-a. After analysis of Table 8 listing statistical information and theoretical wind potential, it can be seen that 42.40 % of Cameroon’s land area is considered unsuitable for large-scale wind power plants. This corresponds mainly to protected areas, land use, transport infrastructure, airports, watercourses, orography and the various criteria based on climatic, orographic, location and watercourses factors in order to optimize the siting of large-scale wind farms. Table 8 also shows that 0.07 % of Cameroon’s land is highly suitable for large-scale wind farms, with a corresponding technical wind potential of around 2 GW in this technical scenario; 8.17 % is most suitable, with a wind potential of around 221.108 GW; the highest percentage, 43.26 %, corresponds to suitability, with a wind potential of 1171.6 GW. 6.10 % of Cameroon’s land area is less suitability, with a wind energy potential of 165.135 GW. Comparing the initial study result with the technical scenario, it can be note that the area corresponding to highly suitability increases from 0 km² to 423,264 km², i.e. an increase of around 423,264 km²; the area corresponding to suitability decreases from 291809.767 km² to 248115.271 km², a decrease of 43694.496 km²; the area corresponding to most suitability increases from 32539.747 km² to 46824.981 km², an increase of 14285.234 km²; the area corresponding to less suitability increases from 5964.161 km² to 34971.342 km², an increase of around 29007.181 km². In addition, wind speed is the main factor influencing this scenario, with a weight of 35 %. In summary, it can be concluded that the transition from the initial scenario to the technical scenario results in an increase in the highly suitability, most suitability and less suitability zones, and a decrease in the unsuitability zone.

The economic scenario (scenario 2) assigns the greatest weight (60.7 %) to economic factors (proximity to power grids, roads and residential areas), followed by technical factors (26.5 %) and social factors (12.8 %). The weights of the various factors corresponding to the economic

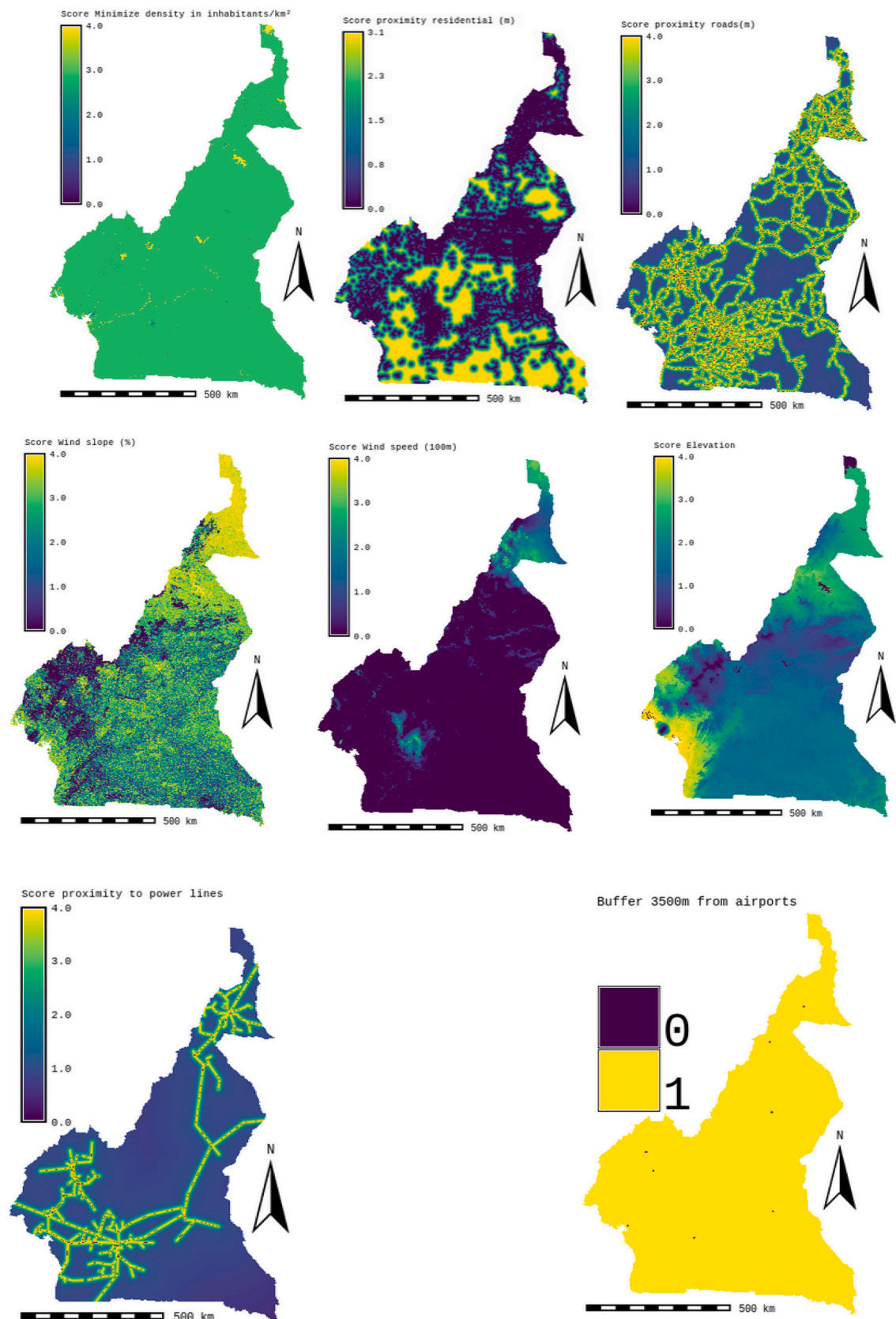


Fig. 5. Score evaluation data.

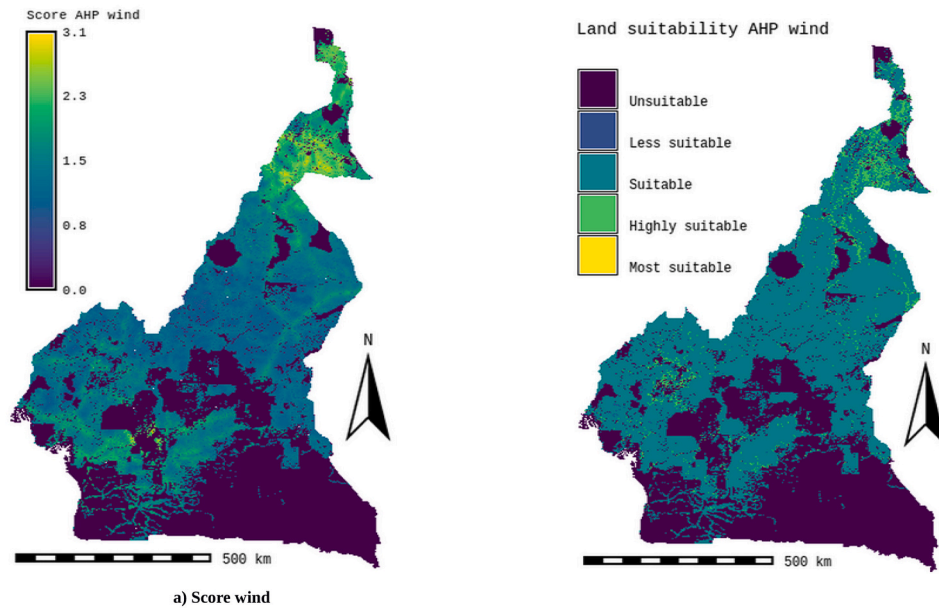


Fig. 6. Score and Land suitability wind.

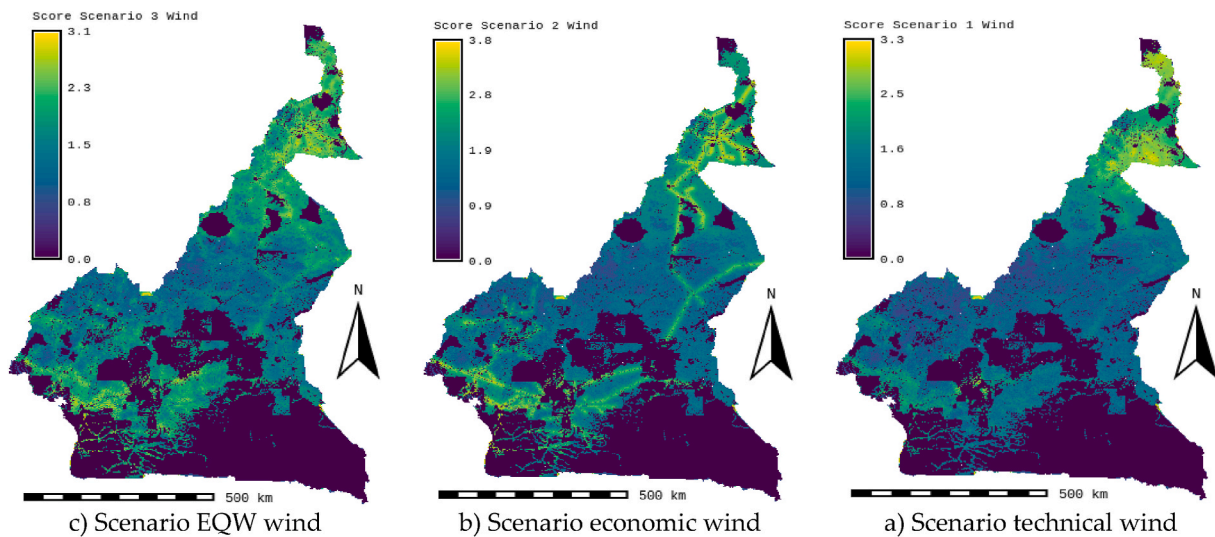


Fig. 7. Score scenarios 1, 2, 3 Wind.

scenario are listed in Table 7, and the suitability maps resulting from the GIS-AHP combination are illustrated in Fig. 5-b and Fig. 6-b. After analysis of Table 8 listing statistical information and wind potential, it can be noted that 42.40 % of Cameroon's land area is considered unsuitable for the siting of large-scale wind power plants. Table 8 also shows that 2.40 % of Cameroonian land is highly suitable for large-scale wind farms, with a corresponding technical wind potential of around 65.093 GW; 18.95 % is most suitable, with a wind potential of around 513.03 GW; 35.58 % and 0.68 % are respectively suitable and less suitable, with wind potentials of 963.384 GW and 18.335 GW respectively. Comparing the initial study result with the socio-economic scenario, it may be noted that the area corresponding to high suitability decreases from 0 km² to 13784.946 km², i.e. an increase of 13784.946 km²; the area corresponding to suitability increases from 291809.767 km² to 204020.342 km², a decrease of 87787.425 km²; the area corresponding to most suitability increases from 32539.747 km² to 108646.7543 km², an increase of 76107.007 km²; the area corresponding to less suitability decreases from 5964.161 km² to 3882.816

km², a decrease of around 2081.345 km². In addition, proximity to the power grid is the main factor influencing this scenario, with a weight of 35 %. Since electrical energy cannot be stored once it has been produced, and since transporting it over long distances leads to losses, it is very economical to build large-scale wind power plants close to power transmission lines, so as to avoid having to build new lines to transport the electricity. This will optimize transmission and reduce investment costs. In summary, it can be concluded that the transition from the initial scenario to the economic scenario results in a decrease in the highly suitability, suitability and less suitability zones; and an increase in the most suitability zones.

The equal weight scenario (scenario 3) assigns the same weight to all factors, i.e. 14 % and 15 % for the 7 factors. This scenario assigns 43 % to technical factors, 43 % to economic factors and 14 % to social factors. The weights of the factors corresponding to the equal-weight scenario are listed in Table 7, and the suitability maps resulting from the GIS-AHP combination are illustrated in Fig. 5d and 6d. After analysis of Table 8 listing the statistical information and theoretical wind potential, it can

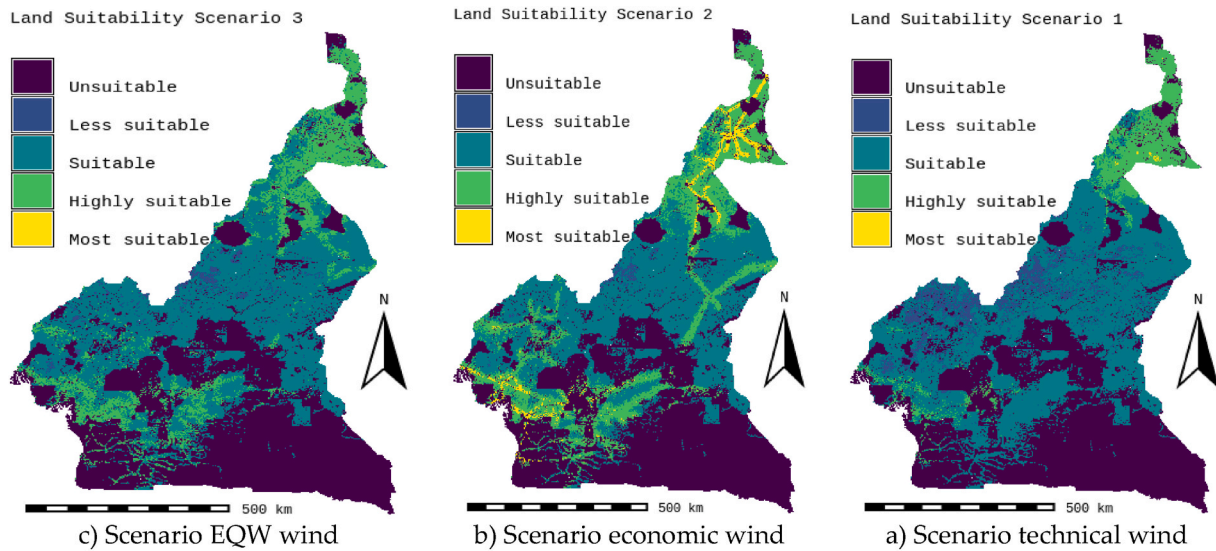


Fig. 8. Land suitability scenario 1, 2, 3.

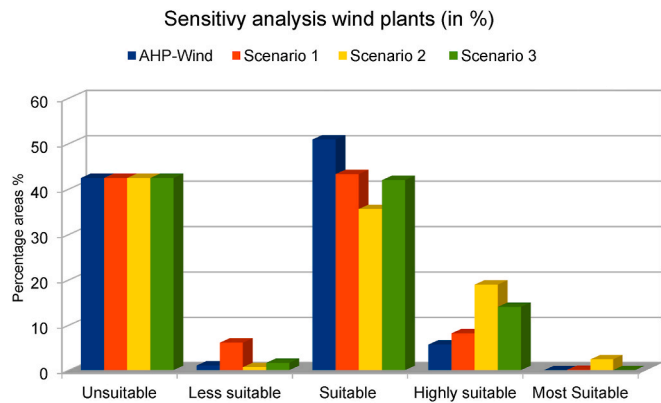


Fig. 9. Graphical interpretation for land suitability areas for sensitivity cases.

note that 42.40 % of Cameroon's land is considered unsuitable for the siting of large-scale wind power plants. The same table shows that 0 % of Cameroon's land is highly suitable for large-scale wind farms, with a corresponding technical wind potential of around 6.91 GW; 14.04 % is most suitable, with a wind potential of around 380.21 GW; 41.94 % and 1.62 % are suitability and less suitability respectively, with wind potentials of 1135.651 GW and 43.975 GW respectively. Comparing the initial result of the study with the equal-weight scenario, it can note that the area corresponding to high suitability increases from 0 km² to 1.29 km², i.e. an increase of around 1.29 km²; the area corresponding to suitability decreases from 291809.767 km² to 240502.028 km², a decrease of 51307.739 km²; the area corresponding to most suitability increases from 32539.747 km² to 80518.772 km², an increase of 47979.025 km²; the area corresponding to less suitability increases from 5964.161 km² to 9312.7672 km², an increase of around 3348.6062 km². It can therefore conclude that the transition from the initial scenario to the equal weight scenario results in a decrease in the suitability zone and an increase in the highly suitability, most suitability and less suitability zones.

The results of the study show that suitable sites are mainly located in the far north of the country, notably in the Extreme North and North regions. There are also suitable sites in the North-West, South-West, West, Littoral and very few in the South. However, the Centre and East regions have no suitable sites. This is mainly due to the tropical forest that covers the whole of East Cameroon, and the low wind speed in these

regions, which is the determining factor for the installation of wind farms.

Table 10 in support to Figs. 10 and 11 presents a disaggregated analysis of different sensitivity cases and the theoretical wind potential of Cameroon's ten regions. It can be seen that the highest potential is found in the Adamaoua region, with a cumulative potential for the different ranges of 333302.37 GW and a total area of the region of 78151 km², followed by the North, Centre, Extreme North, East and South-West regions, with respective potentials of 319084.43 GW, 22436.55 GW, 148813.83 GW, 142646.90 GW and 105621.70 GW for total regional areas of 79414 km², 86845 km², 39934 km², 131453 km² and 32162 km² respectively. It may also be noted that the region with the lowest potential is the Littoral region, with a potential of 49552.67 GW for a total regional area of 59487 km². Despite the South region's considerable surface area of around 59487 km², its potential is the lowest, as its suitable surface area is the smallest at 10494 km².

Furthermore, an analysis of the percentage of suitable range in relation to total range for each region reveals that the lowest percentages are in the South and East regions, with percentages of 17.64 % and 22.98 % respectively. These two regions are mainly covered by forests, which are restrictive zones for wind power plants connected to the existing power grid. In addition, the East is powered mainly by thermal power plants and is not connected to the existing electricity grid. However, the economic factor "proximity to power lines" is an essential one for wind power plants connected to the existing power grid, and has the highest weighting in the economic factors at 15 %. It would therefore make sense to assess the potential of this region for off-grid wind power systems, in order to study the impact of the various constraints on the potential of these regions. The highest percentages are in North-West and Adamaoua, with 92.51 % and 90.32 % respectively. The overall percentage of suitable land is around 57.60 % of the total area, with a suitable surface area of 330295 km² and a total potential of 1559652.99 GW.

Although the wind potential in Cameroon is not ideal due to the variability of this energy source, the combination of solar and wind, or even wind and biomass, could enhance site profitability. The ability of concentrated solar power plants to store thermal energy for times when production is low, along with Cameroon's significant photovoltaic potential, may aid in the optimization of a solar-wind hybrid system. Moreover, the abundant biomass in Cameroon could provide stability and efficiency to a wind-biomass hybrid system. Employing a decision-making method that utilizes fuzzy logic or a probabilistic technique like the Monte Carlo method could further improve the outcomes. These

Table 10
Statistical information and the theoretical wind power potential of land suitability areas for three sensitivity cases.

N°	Regions	Unsuitable			Less suitable			Suitable			High suitable			Most suitable			Total		
		Area Wind in (km ²)	Potential (GW)	Area Wind (km ²)	Potential (GW)	Area Wind (km ²)	Potential (GW)	Area Wind (km ²)	Potential (GW)	Area Wind (km ²)	Potential (GW)	Area Wind (km ²)	Potential (GW)	Area Wind (km ²)	Potential (GW)	Area Wind (km ²)	Potential in GW	suitable Area (km ²)	suitable area (%)
1	Adamaua	7566	18.411.08	3899	18.411.08	6669	314811.0	17	80.27	—	—	—	—	—	—	78151	333.302.37	70585	90.32
2	Central	43974	642.19	136	642.19	40233	189980.2	2502	11.814.44	—	—	—	—	—	—	86845	202.436.86	42871	49.36
3	East	101244	2.450.72	519	2.450.72	29690	140196.2	—	0.00	—	—	—	—	—	—	131453	142.646.90	30209	22.98
4	Extreme-North	8419	0.00	—	0.00	10694	50497.1	20821	98.316.76	—	—	—	—	—	—	39934	148.813.83	31515	78.92
5	Litoral	6673	193.60	41	193.60	16843	79532.6	2436	11.502.79	—	—	—	—	—	—	25993	91.229.04	19320	74.33
6	North	11840	1.298.55	275	1.298.55	61382	289845.8	5917	27.940.07	—	—	—	—	—	—	79414	319.084.43	67574	85.09
7	North-West	1654	2.634.88	558	2.634.88	19834	93656.1	28	132.22	—	—	—	—	—	—	22074	96.423.24	20420	92.51
8	West	2962	467.48	99	467.48	14700	69413.4	140	661.08	—	—	—	—	—	—	17901	70.541.96	14939	83.45
9	South	48993	472.20	100	472.20	10014	47286.1	380	1.794.36	—	—	—	—	—	—	59487	49.552.67	10494	17.64
10	South-West	9794	1.553.54	329	1.553.54	21739	102651.6	300	1.416.60	—	—	—	—	—	—	32162	105.621.70	22368	69.55
Total	243119	5956	291798	28.124.23	291798	1377870.2	32541	153.658.60	—	—	—	—	—	—	—	330295	1559.652.99	1.559.652.99	57.60

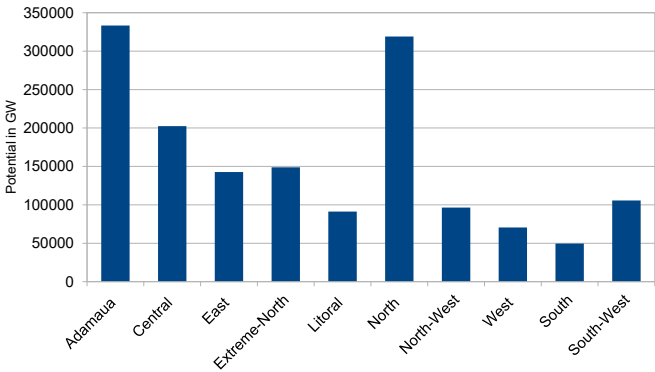


Fig. 10. Power generation potential wind for ten regions of Cameroon.

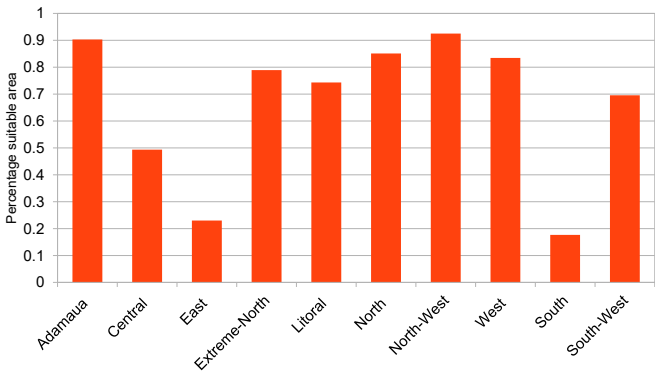


Fig. 11. Percentage of suitable area of wind for ten regions of Cameroon.

topics will be explored in our forthcoming research.

4. Conclusion

This study, which aims to propose potential large-scale wind farm sites using a proposed tool derived from the combination of MCDM-AHP methods coupled with GIS in the Cameroon context, has produced several results. New aspects and factors were taken into account to refine the results obtained. As a result, 57.6 % of Cameroon’s surface area is suitable for onshore wind farms connected to the existing electricity grid. Analysis of the results from the maps developed could help investors, the government and other stakeholders to identify potential areas for the development of wind power systems in the regions of Cameroon. They can also help decision-making on the relative expansion of the power grid to facilitate investment in wind power plants and realize synergies between power generation and transmission. However, it should be noted that the results need to be treated with caution in practice regarding assumptions, data quality, decision-maker preference and the methods applied. For effective implementation, more detailed information and analysis is required. The preference of decision-makers in determining restrictive criteria, evaluation criteria, and their weights have a major impact on results. A sensitivity analysis of the weights was therefore carried out to test the robustness of the results.

This sensitivity analysis also enabled us to decide whether to maximize technical, economic or social factors, or even equiprobability of factors. It can also see from Fig. 9 that the highest percentages in all scenarios are obtained in the second range corresponding to “suitable”, followed by the third range corresponding to “highly suitable”. The lowest percentages are observed in the fourth range corresponding to “most suitable”. An estimate of the theoretical wind potential of large-scale grid-connected wind farms in suitable areas is put at around 1559.741 GW. Although Cameroon does not have enormous wind energy potential across its entire territory, suitable sites for large-scale

wind farms can be found in many parts of the country. Solar-wind hybridization could optimize the output of these power plants, since solar works during the day and wind is optimal at night.

The AHP-based MCDM method used is easy, as it solves a disordered problem by breaking it down into smaller phases. No real data sets are required. AHP thus creates a simple path enabling a non-expert individual to solve complex problems. However, a weakness of the AHP method lies in the use of fundamental qualities for judgments, and for future research in this field. The use of the FUZZY AHP method could make up for the shortcomings observed in the AHP. In addition, the observed results could be improved by introducing other factors such as land cost, lightning flash rate, bird area, appearance, etc. This study was carried out for large-scale onshore wind farms connected to the existing power grid. It would be interesting to carry out a study for off-grid areas in order to propose suitable sites for powering areas not covered by the electricity grid.

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CRediT authorship contribution statement

Isabelle Flora Fotsing Metegam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Isaac Yannick Bomeni:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Venant Sorel Chara-Dackou:** Writing – review & editing, Visualization, Validation, Conceptualization. **Donatien Njomo:** Writing – review & editing, Visualization, Supervision, Conceptualization. **René Tchinda:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest between themselves, nor with any institution or person.

Data availability

Data will be made available on request.

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Nomenclature

MCDM: multi-criteria decision-making
GIS: Geographic Information System
AHP: analytic Hierarchical Process
FAHP: Fuzzy analytic Hierarchy Process
TWPP: theoretical wind power potential
IEA: International Energy Agency
IRENA: International Renewable Energy Agency
PV: photovoltaic
CSP: concentrating solar power
INRA: National Institute of Agricultural Research
CI:: coherence index, consistency index
CR:: consistency rate
RI:: Random Index
A:: area, m²
V: Speed, m.s⁻¹
Z:: height, m
P: the average wind power, W
TA: the total area of suitable land, km²
AF: the area factor in MW/km²
ρ:: density, kg/m³
 $\lambda_{\max}$:: the eigenvalue
n: the number of main criteria
W_i: ith weight of the criterion
P_i: the score of the criterion of the ith factor
S: The total fit score